

#4 Quiz Chapter 11&13

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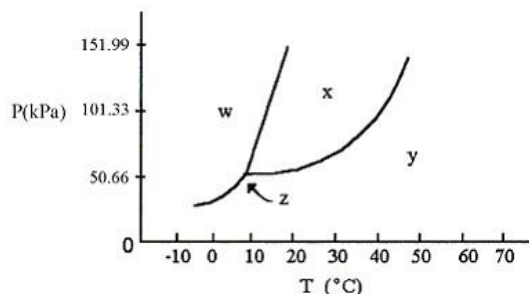
Part A: Multiple choice (Select the one that is best in each case. 1 point/question)

1. As a gaseous element condenses, the atoms become _____ and they have _____ attraction for one another.
A) more separated, more
B) more separated, less
C) closer together, more
D) closer together, less
E) larger, greater
2. When KBr dissolves in water, aqueous K^+ and Br^- ions result. The force of attraction that exists between K^+ and H_2O is called a(n) _____ interaction.
A) dipole-dipole
B) ion-dipole
C) ion-ion
D) London dispersion force
E) hydrogen bonding
3. The intermolecular force(s) responsible for the fact that CH_4 has the lowest boiling point in the set CH_4 , CH_3CH_3 , $CH_3CH_2CH_3$, $CH_3CH_2CH_2CH_3$ is/are _____.
A) hydrogen bonding
B) London dispersion forces
C) mainly hydrogen bonding but also dipole-dipole forces
D) dipole-dipole forces
E) mainly dipole-dipole forces but also London dispersion forces
4. Which one of the following substances will have hydrogen bonding as one of its intermolecular forces?
A) formaldehyde, H_2CO
B) phosphine, PH_3
C) chloramine, NH_2Cl
D) hydrogen sulfide, H_2S
E) fluoromethane, CH_3F
5. What types of intermolecular forces exist between water and HF?
A) dispersion forces and dipole-dipole forces
B) dispersion forces, dipole-dipole forces, and hydrogen bonds
C) dispersion forces and hydrogen bonds
D) dispersion forces
E) dispersion forces, hydrogen bonds, and ion-dipole forces
6. The substance with the largest heat of vaporization is _____.
A) H_2 B) Cl_2 C) I_2 D) N_2 E) O_2

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7. According to the phase diagram shown below, what is the normal melting point (°C)?



- A) 14 B) 25 C) 35 D) 40 E) 45

8. What information about water is needed to calculate the enthalpy change for converting 1 mol $\text{H}_2\text{O}(s)$ at 0 °C to $\text{H}_2\text{O}(l)$ at 80 °C?

- A) heat of fusion
B) specific heat of $\text{H}_2\text{O}(l)$
C) heat of vaporization and specific heat of $\text{H}_2\text{O}(l)$
D) heat of fusion and specific heat of $\text{H}_2\text{O}(l)$
E) heat of fusion and specific heat of $\text{H}_2\text{O}(s)$

9. Of the following, _____ is the most volatile.

- A) C_2H_6 B) C_2I_6 C) C_2Br_6 D) C_2Cl_6 E) C_2F_6

10. Liquids form a curved surface called a *meniscus* when placed in a glass tube. Which statement is true about the meniscus of mercury?

- A) It is U-shaped because the adhesive forces are greater than the cohesive forces.
B) It is shaped like an inverted U because the adhesive forces are stronger than the cohesive forces.
C) It is shaped like an inverted U because the cohesive forces are stronger than the adhesive forces.
D) It is U-shaped because the cohesive forces are stronger than the adhesive forces.
E) None of these is correct.

11. The highest temperature at which a substance can exist in its liquid state is called its _____ point.

- A) Boiling B) freezing C) triple D) critical

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12. Consider two ionic solids, both composed of singly-charged ions, that have different lattice energies. which solid will be more soluble in water? Assume that solute-solvent interactions are the same for both solids.
- A) The solids will have the same solubility in water
 - B) The one with the larger lattice energy
 - C) The one with the smaller lattice energy
13. Colligative properties of solutions include all of the following except _____.
- A) depression of vapor pressure upon addition of a solute to a solvent
 - B) elevation of the boiling point of a solution upon addition of a solute to a solvent
 - C) depression of the freezing point of a solution upon addition of a solute to a solvent
 - D) an increase in the osmotic pressure of a solution upon the addition of more solute
 - E) the increase of reaction rates with increase in temperature
14. The phrase "like dissolves like" refers to the fact that _____.
- A) gases can only dissolve other gases
 - B) Substances with similar intermolecular attractive forces tend to be soluble in one another
 - C) solvents can only dissolve solutes of similar molar mass
 - D) condensed phases can only dissolve other condensed phases
 - E) polar solvents dissolve nonpolar solutes and vice versa
15. 25.0 g of KBr is dissolved in enough water to produce 456 mL of the solution. What is the molarity of this solution?
- A) 0.210 M B) 54.8 M C) 0.461 M D) 0.921 M E) $4.61 \times 10^{-4} M$
16. What is the molecular weight of adrenaline if 0.64 grams of adrenaline dissolved in 36.0 g of CCl₄ raises the boiling point by 0.49 degrees Celsius (K_b for CCl₄ = 5.02 C/molal)?
- A) 180 g/mole B) 360 g/mole C) 720 g/mole D) 1800 g/mole
17. What is the freezing point of a 2.00 m aqueous solution of sucrose? The value of K_f for water is 1.86 degrees C/molal?
- A) -3.72 degrees C; B) -1.86 degrees C; C) +1.86 degrees C; D) +3.72 degrees C

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18. The average rate of disappearance of ozone in the reaction $2\text{O}_3(\text{g}) \rightarrow 3\text{O}_2(\text{g})$ is found to be 8.8×10^{-3} atm over a certain interval of time. What is the rate of appearance of O_2 during this interval?

- A) 5.9×10^{-3} atm/time
- B) 2.6×10^{-2} atm/time
- C) 1.8×10^{-2} atm/time
- D) 1.3×10^{-2} atm/time
- E) 8.8×10^{-3} atm/time

1-5 CBBCB

6-10 CADAC

11-15 DCEBC

16-18 AAD

Part B: Short Answer (Write legibly and show all work for all steps in the problem)

19. As the intermolecular attractive forces between molecules increase in magnitude, do you expect each of the following to increase or decrease in magnitude? (a) vapor pressure, (b) heat of vaporization, (c) boiling point, (d) freezing point, (e) viscosity, (f) surface tension, (g) critical temperature. (4 points)

(a) decrease (b) increase (c) increase (d) increase (e) increase (f) increase (g) increase

20. How much energy is needed to convert 60.6 grams of ice at 0.00°C to water at 55.4°C ? (2 points)

Specific heat of ice = $2.10 \text{ J}/(\text{g } ^\circ\text{C})$

Specific heat of water = $4.18 \text{ J}/(\text{g } ^\circ\text{C})$

Heat of fusion = 333 J/g

Heat of vaporization = 2258 J/g

$$E = 333 \text{ J/g} \times 60.6 \text{ g} + 55.4 ^\circ\text{C} \times 4.18 \text{ J}/(\text{g } ^\circ\text{C}) \times 60.6 \text{ g} = 34213 \text{ J} = 34.21 \text{ kJ}$$

21. List the substances BaCl_2 , H_2 , CO , HF , and Ne in order of increasing boiling point. (2 points)

$\text{H}_2 < \text{Ne} < \text{CO} < \text{HF} < \text{BaCl}_2$

22. Predict whether each of the following substances is more likely to dissolve in the nonpolar solvent carbon tetrachloride CCl_4 or in water: C_7H_{16} , Na_2SO_4 , HCl , and I_2 . (2 points)

CCl_4 : C_7H_{16} , I_2

H_2O : Na_2SO_4 , HCl

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23. Calculate the concentration of CO_2 in a soft drink that is bottled with a partial pressure of CO_2 of 4.0 atm over the liquid at 25 °C. The Henry's law constant for CO_2 in water at this temperature is $3.4 \times 10^{-2} \text{ mol/L-atm}$. (2 points)

Solve $S_{\text{CO}_2} = kP_{\text{CO}_2} = (3.4 \times 10^{-2} \text{ mol/L-atm})(4.0 \text{ atm}) = 0.14 \text{ mol/L} = 0.14 M$

24. A solution is made by dissolving 4.35 g of glucose $\text{C}_6\text{H}_{12}\text{O}_6$ in 25.0 mL of water at 25 °C. Calculate the molality of glucose in the solution. Water has a density of 1.00 g/mL. (4 points)

Solve Use the molar mass of glucose, 180.2 g/mol, to convert grams to moles:

$$\text{Mol } \text{C}_6\text{H}_{12}\text{O}_6 = (4.35 \text{ g } \text{C}_6\text{H}_{12}\text{O}_6) \left(\frac{1 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6}{180.2 \text{ g } \text{C}_6\text{H}_{12}\text{O}_6} \right) = 0.0241 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6$$

Because water has a density of 1.00 g/mL, the mass of the solvent is

$$(25.0 \text{ mL})(1.00 \text{ g/mL}) = 25.0 \text{ g} = 0.0250 \text{ kg}$$

Finally, use Equation 13.9 to obtain the molality:

$$\text{Molality of } \text{C}_6\text{H}_{12}\text{O}_6 = \frac{0.0241 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6}{0.0250 \text{ kg } \text{H}_2\text{O}} = 0.964 m$$

25. An aqueous solution of hydrochloric acid contains 36% HCl by mass. (a) Calculate the mole fraction of HCl in the solution. (b) Calculate the molality of HCl in the solution. (4 points)

Solve (a) To calculate the mole fraction of HCl, we convert the masses of HCl and H_2O to moles and then use Equation 13.7:

$$\text{Moles HCl} = (36 \text{ g HCl}) \left(\frac{1 \text{ mol HCl}}{36.5 \text{ g HCl}} \right) = 0.99 \text{ mol HCl}$$

$$\text{Moles } \text{H}_2\text{O} = (64 \text{ g } \text{H}_2\text{O}) \left(\frac{1 \text{ mol } \text{H}_2\text{O}}{18 \text{ g } \text{H}_2\text{O}} \right) = 3.6 \text{ mol } \text{H}_2\text{O}$$

$$X_{\text{HCl}} = \frac{\text{moles HCl}}{\text{moles } \text{H}_2\text{O} + \text{moles HCl}} = \frac{0.99}{3.6 + 0.99} = \frac{0.99}{4.6} = 0.22$$

(b) To calculate the molality of HCl in the solution, we use Equation 13.9. We calculated the number of moles of HCl in part (a), and the mass of solvent is 64 g = 0.064 kg:

$$\text{Molality of HCl} = \frac{0.99 \text{ mol HCl}}{0.064 \text{ kg } \text{H}_2\text{O}} = 15 m$$

Notice that we can't readily calculate the molarity of the solution because we don't know the volume of the 100 g of solution.

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26. Use the phase diagram for methane, CH₄, shown in the Figure below to answer the following questions. (a) What are the approximate temperature and pressure of the critical point? (b) What are the approximate temperature and pressure of the triple point? (c) Is methane a solid, liquid, or gas at 1 atm and 0 °C? (d) If solid methane at 1 atm is heated while the pressure is held constant, will it melt or sublime? (e) If methane at 1 atm and 0 °C is compressed until a phase change occurs, in which state is the methane when the compression is complete? (5 points)

(a) The critical point is the point where the liquid, gaseous, and supercritical fluid phases coexist. It is marked point 3 in the phase diagram and located at approximately -80 °C and 50 atm.

(b) The triple point is the point where the solid, liquid, and gaseous phases coexist. It is marked point 1 in the phase diagram and located at approximately -180 °C and 0.1 atm.

(c) The intersection of 0 °C and 1 atm is marked point 2 in the phase diagram. It is well within the gaseous region of the phase diagram.

(d) If we start in the solid region at P = 1 atm and move horizontally (this means we hold the pressure constant), we cross first into the liquid region, at T ≈ -180 °C, and then into the gaseous region, at T ≈ -160 °C. Therefore, solid methane melts when the pressure is 1 atm. (For methane to sublime, the pressure must be below the triple point pressure.)

(e) Moving vertically up from point 2, which is 1 atm and 0 °C, the first phase change we come to is from gas to supercritical fluid. This phase change happens when we exceed the critical pressure (~50 atm).

27. This portion of a phase diagram shows the vapor-pressure curves of a volatile solvent and of a solution of that solvent containing a nonvolatile solute. (a) Which line represents the solution? (b) What are the normal boiling points of the solution? (2 points)

(a) The blue line represents the solution.

(b) 70 °C

28. (a) At 20 °C, the vapor pressure of benzene (C₆H₆) is 75 torr, and that of toluene (C₇H₈) is 22 torr. Assume that benzene and toluene form an ideal solution. What is the composition in mole fraction of a solution that has a vapor pressure of 35 torr at 20 °C? (b) What is the mole fraction of benzene in the vapor above the solution described in part (a)? (4 points)

(a) Benzene and toluene form an ideal solution.

$$P = x_{\text{C}_6\text{H}_6} P_{\text{C}_6\text{H}_6}^{\circ} + x_{\text{C}_7\text{H}_8} P_{\text{C}_7\text{H}_8}^{\circ} = 75x_{\text{C}_6\text{H}_6} + 22(1 - x_{\text{C}_6\text{H}_6}) = 35$$

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$$x_{\text{C}_6\text{H}_6} = 0.25$$

$$x_{\text{C}_7\text{H}_8} = 1 - 0.245 = 0.75$$

$$(b) P_{\text{C}_6\text{H}_6} = x_{\text{C}_6\text{H}_6} P_{\text{C}_6\text{H}_6}^{\circ} = 0.2453 \times 75 = 18.4 \text{ torr}$$

$$\text{mole fraction of C}_6\text{H}_6 = \frac{P_{\text{C}_6\text{H}_6}}{P_t} = \frac{18.4 \text{ torr}}{35 \text{ torr}} = 0.53$$

29. Glycerin $\text{C}_3\text{H}_8\text{O}_3$ is a nonvolatile nonelectrolyte with a density of 1.26 g/mL at 25 °C. Calculate the vapor pressure at 25 °C of a solution made by adding 50.0 mL of glycerin to 500.0 mL of water. The vapor pressure of pure water at 25 °C is 23.8 torr, and its density is 1.00 g/mL. (4 points)

Solve To calculate the mole fraction of water in the solution, we must determine the number of moles of $\text{C}_3\text{H}_8\text{O}_3$ and H_2O :

$$\text{Moles C}_3\text{H}_8\text{O}_3 = (50.0 \text{ mL C}_3\text{H}_8\text{O}_3) \left(\frac{1.26 \text{ g C}_3\text{H}_8\text{O}_3}{1 \text{ mL C}_3\text{H}_8\text{O}_3} \right) \left(\frac{1 \text{ mol C}_3\text{H}_8\text{O}_3}{92.1 \text{ g C}_3\text{H}_8\text{O}_3} \right) = 0.684 \text{ mol}$$

$$\text{Moles H}_2\text{O} = (500.0 \text{ mL H}_2\text{O}) \left(\frac{1.00 \text{ g H}_2\text{O}}{1 \text{ mL H}_2\text{O}} \right) \left(\frac{1 \text{ mol H}_2\text{O}}{18.0 \text{ g H}_2\text{O}} \right) = 27.8 \text{ mol}$$

$$X_{\text{H}_2\text{O}} = \frac{\text{mol H}_2\text{O}}{\text{mol H}_2\text{O} + \text{mol C}_3\text{H}_8\text{O}_3} = \frac{27.8}{27.8 + 0.684} = 0.976$$

We now use Raoult's law to calculate the vapor pressure of water for the solution:

$$P_{\text{H}_2\text{O}} = X_{\text{H}_2\text{O}} P_{\text{H}_2\text{O}}^{\circ} = (0.976)(23.8 \text{ torr}) = 23.2 \text{ torr}$$

30. Automotive antifreeze contains ethylene glycol, $\text{CH}_2(\text{OH})\text{CH}_2(\text{OH})$, a nonvolatile nonelectrolyte, in water. Calculate the boiling point and freezing point of a 25.0% by mass solution of ethylene glycol in water. For pure water, $K_b = 0.51 \text{ }^{\circ}\text{C}/m$, $K_f = 1.86 \text{ }^{\circ}\text{C}/m$ (4 points)

Solve The molality of the solution is calculated as follows:

$$\begin{aligned} \text{Molality} &= \frac{\text{moles C}_2\text{H}_6\text{O}_2}{\text{kilograms H}_2\text{O}} = \left(\frac{250 \text{ g C}_2\text{H}_6\text{O}_2}{750 \text{ g H}_2\text{O}} \right) \left(\frac{1 \text{ mol C}_2\text{H}_6\text{O}_2}{62.1 \text{ g C}_2\text{H}_6\text{O}_2} \right) \left(\frac{1000 \text{ g H}_2\text{O}}{1 \text{ kg H}_2\text{O}} \right) \\ &= 5.37 m \end{aligned}$$

We can now use Equations 13.12 and 13.13 to calculate the changes in the boiling and freezing points:

$$\Delta T_b = iK_b m = (1)(0.51 \text{ }^{\circ}\text{C}/m)(5.37 m) = 2.7 \text{ }^{\circ}\text{C}$$

$$\Delta T_f = -iK_f m = -(1)(1.86 \text{ }^{\circ}\text{C}/m)(5.37 m) = -10.0 \text{ }^{\circ}\text{C}$$

Hence, the boiling and freezing points of the solution are readily calculated:

$$\Delta T_b = T_b(\text{solution}) - T_b(\text{solvent})$$

$$2.7 \text{ }^{\circ}\text{C} = T_b(\text{solution}) - 100.0 \text{ }^{\circ}\text{C}$$

$$T_b(\text{solution}) = 102.7 \text{ }^{\circ}\text{C}$$

$$\Delta T_f = T_f(\text{solution}) - T_f(\text{solvent})$$

$$-10.0 \text{ }^{\circ}\text{C} = T_f(\text{solution}) - 0.0 \text{ }^{\circ}\text{C}$$

$$T_f(\text{solution}) = -10.0 \text{ }^{\circ}\text{C}$$

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31. A solution of an unknown nonvolatile nonelectrolyte was prepared by dissolving 0.250 g of the substance in 40.0 g of CCl_4 . The boiling point of the resultant solution was 0.357°C higher than that of the pure solvent. Calculate the molar mass of the solute. For pure CCl_4 , $K_b = 5.02^\circ\text{C}/m$. (4 points)

Solve From Equation 13.12, we have

$$\text{Molality} = \frac{\Delta T_b}{iK_b} = \frac{0.357^\circ\text{C}}{(1)5.02^\circ\text{C}/m} = 0.0711 m$$

Thus, the solution contains 0.0711 mol of solute per kilogram of solvent. The solution was prepared using $40.0 \text{ g} = 0.0400 \text{ kg}$ of solvent (CCl_4). The number of moles of solute in the solution is therefore

$$(0.0400 \text{ kg CCl}_4) \left(0.0711 \frac{\text{mol solute}}{\text{kg CCl}_4} \right) = 2.84 \times 10^{-3} \text{ mol solute}$$

The molar mass of the solute is the number of grams per mole of the substance:

$$\text{Molar mass} = \frac{0.250 \text{ g}}{2.84 \times 10^{-3} \text{ mol}} = 88.0 \text{ g/mol}$$

32. The osmotic pressure of an aqueous solution of a certain protein was measured to determine the protein's molar mass. The solution contained 3.50 mg of protein dissolved in sufficient water to form 5.00 mL of solution. The osmotic pressure of the solution at 25°C was found to be 1.54 torr. Treating the protein as a nonelectrolyte, calculate its molar mass. $R = 0.0821 \text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K})$, $1 \text{ atm} = 760 \text{ torr}$. (4 points)

Solve Solving Equation 13.14 for molarity gives

$$\text{Molarity} = \frac{\Pi}{iRT} = \frac{(1.54 \text{ torr}) \left(\frac{1 \text{ atm}}{760 \text{ torr}} \right)}{(1) \left(0.0821 \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}} \right) (298 \text{ K})} = 8.28 \times 10^{-5} \frac{\text{mol}}{\text{L}}$$

Because the volume of the solution is $5.00 \text{ mL} = 5.00 \times 10^{-3} \text{ L}$, the number of moles of protein must be

$$\text{Moles} = (8.28 \times 10^{-5} \text{ mol/L}) (5.00 \times 10^{-3} \text{ L}) = 4.14 \times 10^{-7} \text{ mol}$$

The molar mass is the number of grams per mole of the substance. Because we know the sample has a mass of $3.50 \text{ mg} = 3.50 \times 10^{-3} \text{ g}$, we can calculate the molar mass by dividing the number of grams in the sample by the number of moles we just calculated:

$$\text{Molar mass} = \frac{\text{grams}}{\text{moles}} = \frac{3.50 \times 10^{-3} \text{ g}}{4.14 \times 10^{-7} \text{ mol}} = 8.45 \times 10^3 \text{ g/mol}$$